

UTSHO DAS

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Education

Shahjalal University of Science and Technology

2019 – 2025

B.Sc. in Computer Science & Engineering

CGPA: 3.71/4.00

- **Final Two Years GPA:** 3.85/4.00 (75.5 credits)

Standardized Tests

GRE: 326 (Quant: 167, Verbal: 159, AWA: 3.5)

Research Interests

Usable Security & Privacy, Cybersecurity Education, User Mental Models of Autonomous Systems, AI Risk Perception, and Trustworthy Machine Learning.

Research & Publications

CyQured: Design, Development, and Empirical Evaluation of a Tabletop Game for Personal Cybersecurity Education

- Accepted at the **Twenty-Second Symposium on Usable Privacy and Security (SOUPS 2026)**.
- Designed and empirically evaluated a tabletop game to improve personal cybersecurity awareness through interactive gameplay involving digital assets, cyberattacks, and defense mechanisms.
- [Paper Link](#)

A Deep Learning Approach to Automate Classification of Arsenic-Affected Skin using EfficientNet-B1

- Achieved 94.69% accuracy in arsenic-affected skin classification using EfficientNet-B1, outperforming VGG-19, ResNet-50, and MobileNet-v2.
- Presented at ICCIT 2024, Cox's Bazar, Bangladesh.
- [Paper Link](#)

Project PRIMUS at EHRSQL 2024: Text-to-SQL Generation using Large Language Model for EHR Analysis

- Developed a transformer-based Text-to-SQL system using SQLCoders with schema-aware prompting for Electronic Health Record analysis.
- Published in the 6th Clinical NLP Workshop, Mexico City, Mexico.
- [Paper Link](#)

What Can It Really Do? Understanding End-User Mental Models of AI Agent Permissions Across Everyday Platforms (*Ongoing Research*)

- Investigating how end-users understand AI agent permissions across messaging, productivity, and financial platforms using interviews, think-aloud studies, and card-sorting tasks.
- Examines users' mental models of AI autonomy and derives design recommendations for safer AI permission interfaces for non-expert users.

Work Experience

Therap(BD) Ltd.

July 2025 – Present

Associate Database Engineer

Dhaka, Bangladesh

- Develop Oracle SQL solutions and administer enterprise healthcare databases through performance optimization, patch management, backup and recovery, and database maintenance.
- Implement and maintain database security controls including Oracle Database Vault, role-based access control, privileged access restrictions, auditing, and database hardening to protect critical healthcare systems.
- Support HIPAA compliance by safeguarding Protected Health Information (PHI) through secure database practices, least-privilege access, and enforcement of organizational security policies.

Projects

E-Shop

- MERN stack-based e-commerce platform with APIs for payment, product management, and inventory.
- [GitHub](#)

ServiceHub

- Platform built using EJS, JavaScript, MySQL for connecting service providers and consumers.
- [GitHub](#)

Ride-with-the-wave

- Endless runner game in Java using JavaFX for GUI and controls.
- [GitHub](#)

Home Automation System

- Arduino + ESP32 based smart automation and security for homes.
- [Demo](#)

Research & Technical Skills

HCI & User Research: User Study Design, Semi-Structured Interviews, Think-Aloud Studies, Card Sorting, Usability Evaluation

Research Methods: Mixed-Methods Research, Survey Research, Qualitative and Quantitative Analysis, Thematic Analysis

Research Tools: Python, Jupyter Notebook, pandas, NumPy, scikit-learn

Programming: Python, Java, C++, JavaScript, Dart, $\text{\LaTeX}$

Web & Backend: React, Node.js, Express.js, HTML, CSS, Bootstrap

Databases & Cloud: Oracle, MySQL, MongoDB, AWS

Development Tools: Git, GitHub, Docker, VS Code, Postman

Problem Solving Skills

Solved 700+ problems across platforms like Codeforces, LeetCode, LightOJ, and UVa.